

Experiment 3 : write the python program for water jug problem

Aim

To solve the Water Jug Problem using Python and obtain the target quantity of water using two jugs of different capacities.

Algorithm

1. Fill the first jug completely.
2. Pour water from the first jug into the second jug until the second jug becomes full.
3. Empty the second jug when it is full.
4. Continue transferring water between the jugs.
5. Repeat the process until the target quantity is obtained.

Program

```
# Water Jug Problem
```

```
jug1 = 4
```

```
jug2 = 3
```

```
target = 2
```

```
print("Jug1\tJug2")
```

```
a = 0
```

```
b = 0
```

```
while a != target and b != target:
```

```
    if a == 0:
```

```
        a = jug1
```

```
        print(a, "\t", b)
```

```
    elif b == jug2:
```

```
        b = 0
```

```
        print(a, "\t", b)
```

```
    else:
```

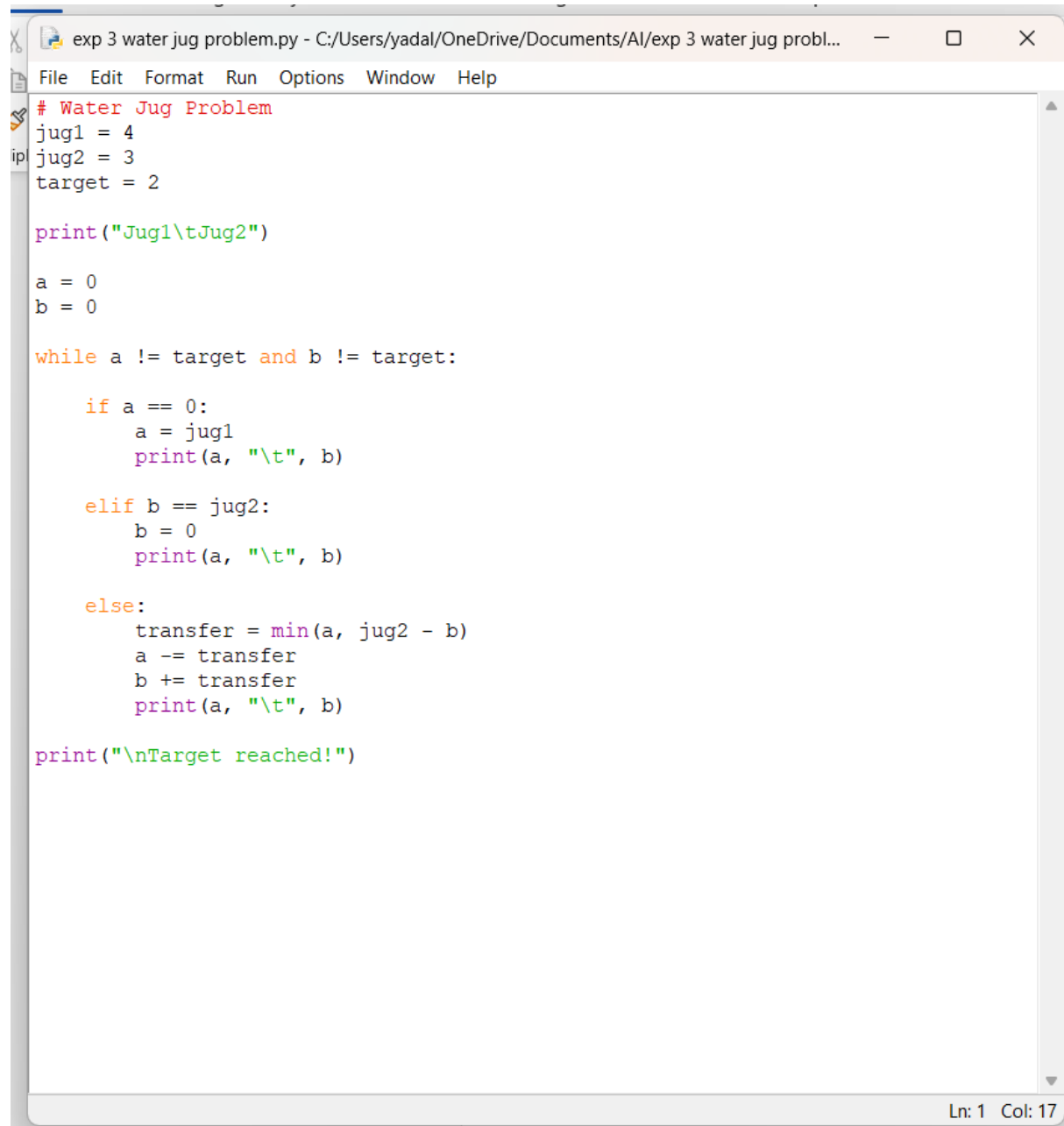
```
        transfer = min(a, jug2 - b)
```

```
        a -= transfer
```

```
b += transfer

print(a, "\t", b)

print("\nTarget reached!")
```

A screenshot of a Python IDE window titled "exp 3 water jug problem.py". The window contains the following Python code:

```
# Water Jug Problem
jug1 = 4
jug2 = 3
target = 2

print("Jug1\tJug2")

a = 0
b = 0

while a != target and b != target:

    if a == 0:
        a = jug1
        print(a, "\t", b)

    elif b == jug2:
        b = 0
        print(a, "\t", b)

    else:
        transfer = min(a, jug2 - b)
        a -= transfer
        b += transfer
        print(a, "\t", b)

print("\nTarget reached!")
```

The status bar at the bottom right indicates "Ln: 1 Col: 17".

Sample Output

```
Jug1  Jug2

4    0

1    3

1    0
```

0 1

4 1

2 3

Target reached!

```
>>> ~
=== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 3 water jug problem.py ===
Jug1    Jug2
4        0
1        3
1        0
0        1
4        1
2        3
Target reached!
>>>
```

Result

The Water Jug Problem was successfully solved using Python. The target quantity of **2 liters** was obtained using a **4-liter jug** and a **3-liter jug**.